Writing the methods section

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Writing the methods section before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Figures that preserve evidence

Without opening the old lesson, explain one key idea from Figures that preserve **evidence**. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Robustness across conditions

Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Reproducible environment and records

Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Writing the methods section trace and record the first step you can explain without help.

Use W2: Trace the worked case for Writing the **methods section**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Writing the methods section explanation against the supplied facts before discussing it.

Use the worked case for Writing the **methods section**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

> Methods names a dataset and metric, but not whether near-duplicate sources cross the split. What detail matters?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Writing the methods section facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Writing the methods section facts and keep your reasoning.

Methods says used AI on data and got good scores; code and split are absent. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Writing the methods section facts and keep your reasoning.

A report omits this qualification: A detailed **methods section** can reveal limitations; it does not guarantee a correct design. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Writing the methods section facts and keep your reasoning.

Methods names a dataset and metric, but not whether near-duplicate sources cross the split. What detail matters?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

Design a revealing follow-up. Apply the supplied Writing the methods section facts and keep

> your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Writing the methods section to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled writing the **methods section evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of writing the **methods section**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Writing the methods section

> Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule. Explain the labelled step before checking the changed condition.

Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule.

Inputs: "oak", "pine", "willow". Rule: length at least 4→long, otherwise short. No whitespace trimming occurs.

AI ACADEMY / AI-7/30

Writing the methods section

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record exact inputs

Preserve the three strings and no-trimming policy.



2. State procedure

Apply the inclusive length boundary at four characters.



3. Show resulting labels

Oak is short; pine and willow are long.

CHANGE ONE THING

Change the rule to length greater than 4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Apply the inclusive four-character rule to oak, pine and willow. Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Oak is short; pine and willow are long.

1. State procedure

2. Record exact **input**s

3. Show resulting labels

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Show resulting labels Oak is short; pine and willow are long. A **methods section** specifies **input**s, **process**ing, boundaries, versions and scoring.

Precise procedures allow a reader to repeat the computation instead of inferring it from the final labels.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change the rule to length greater than 4. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Pine becomes short; willow stays long and oak stays short.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Pine becomes short; willow

stays long and oak stays short. The equality case distinguishes the two procedures.

Repair a missing building block

Review Figures that preserve **evidence** (ai-7/29). Goal: Explain the **mechanism** and terminology of figures that preserve **evidence**. Short practice: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?. Return to this week and explain what you changed.

Review Robustness across conditions (ai-7/27). Goal: Explain the **mechanism** and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates.

Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Review Reproducible environment and records (ai-7/24). Goal: Explain the **mechanism** and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9.

Which artefacts must accompany the **output** so another team can identify the exact run?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

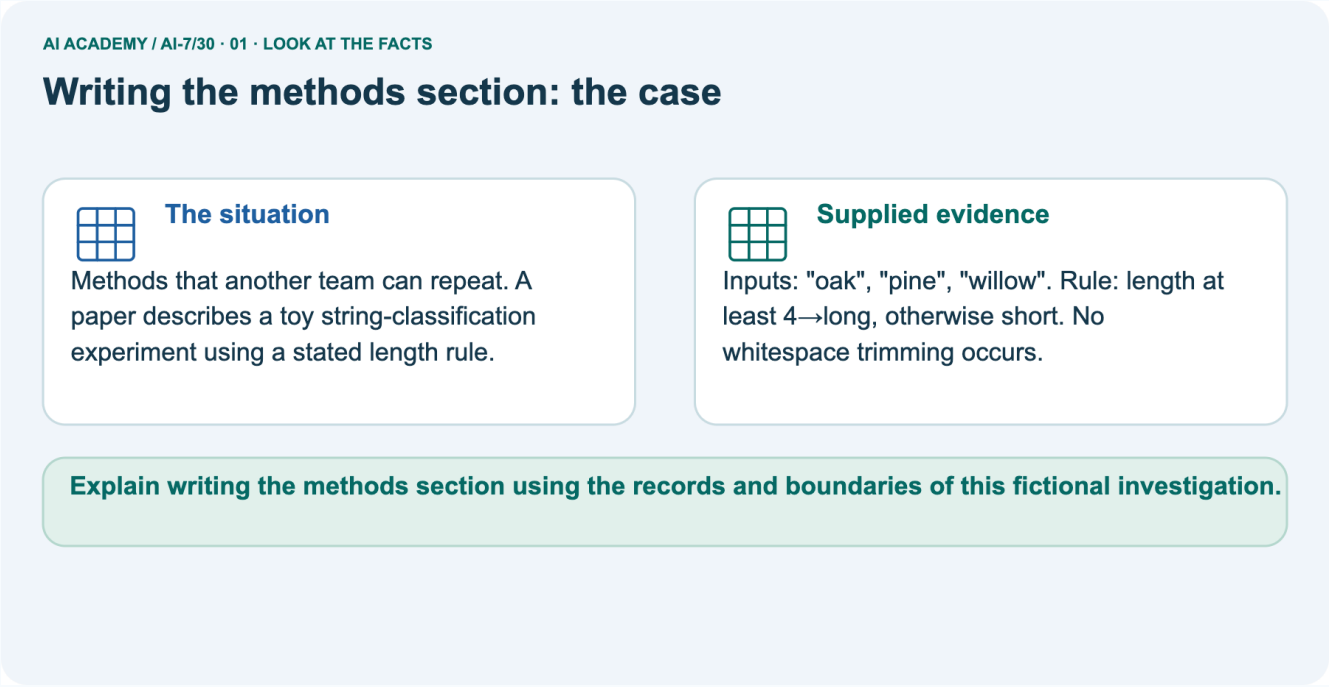
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Writing the methods section picture from its labels, then record a relevant limit.

01. Writing the methods section: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

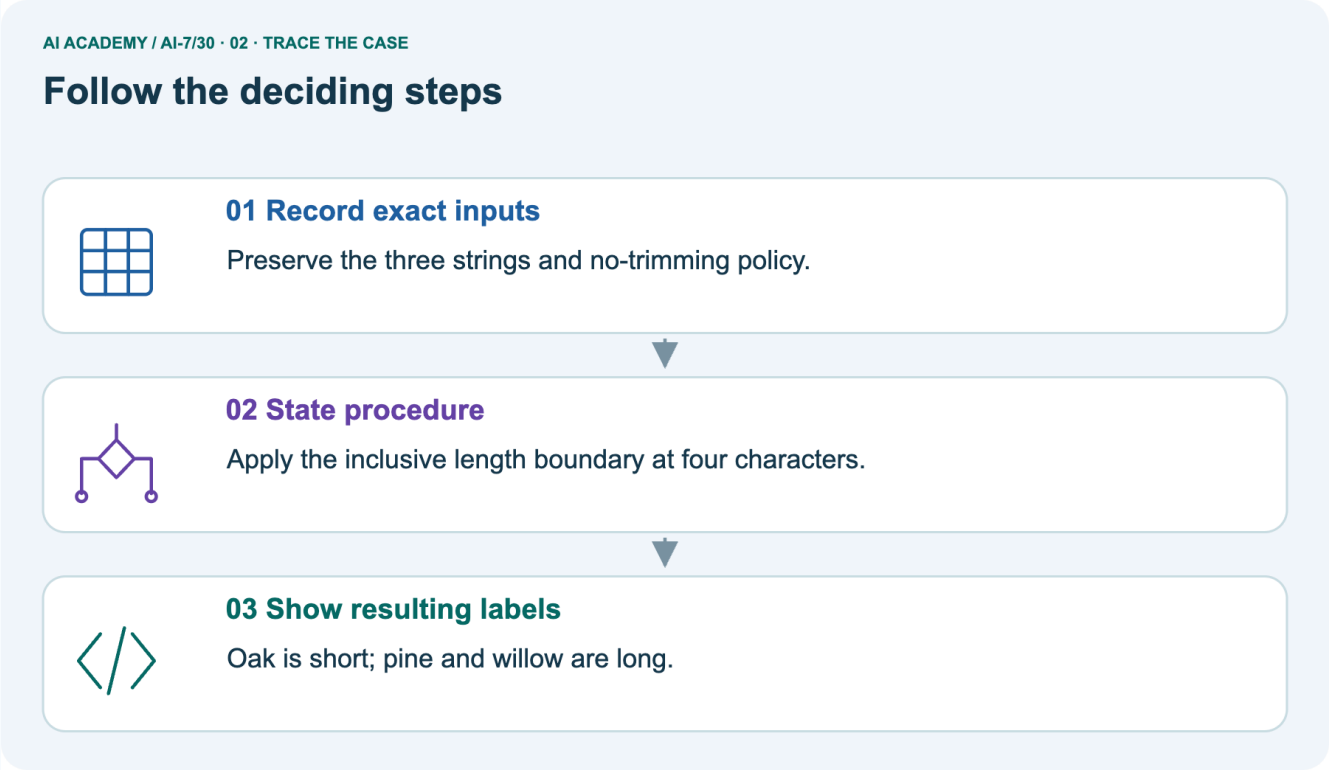
Methods that another team can repeat. A paper describes a toy string-classification experiment using a stated length rule. Supplied facts: **Input**s: "oak", "pine", "willow". Rule: length at least 4→long, otherwise short.

No whitespace trimming occurs.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



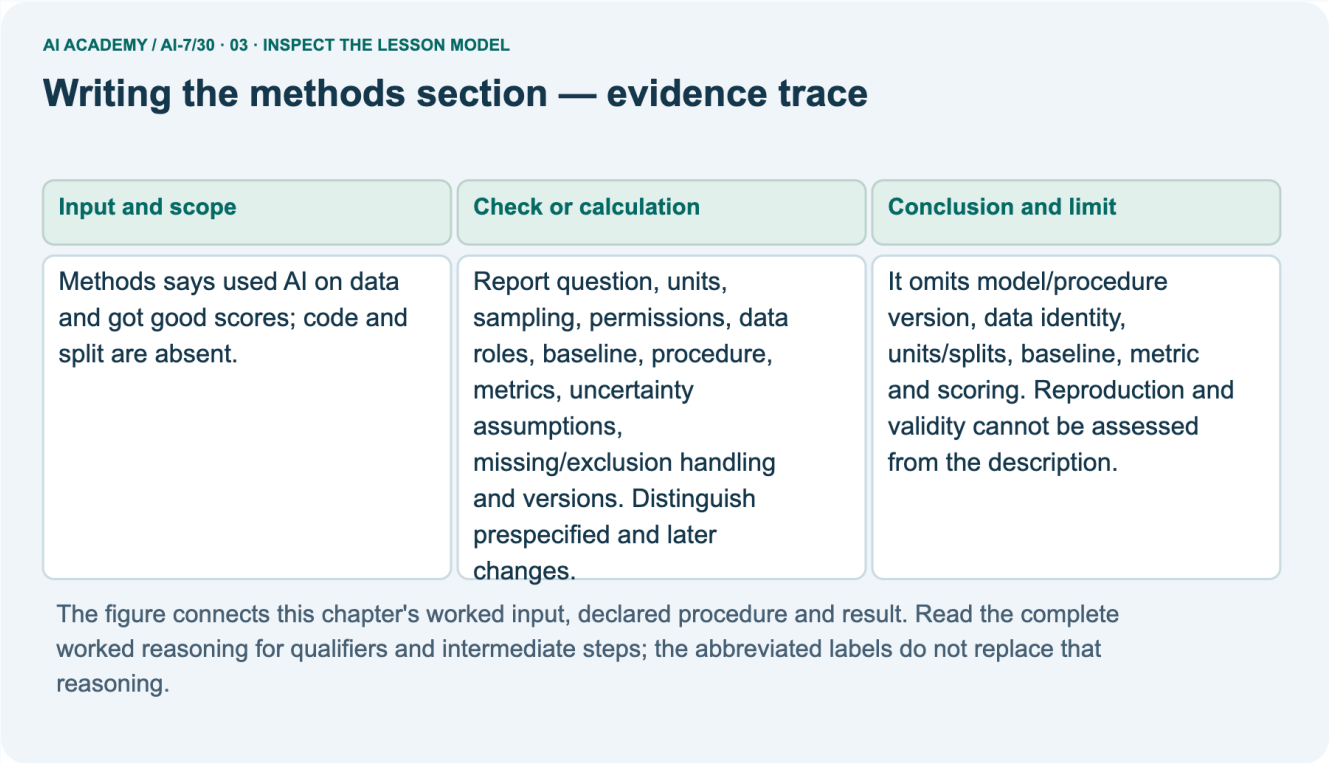
Follow the labelled picture with the matching explanation. Open the vector diagram

Record exact **input**s: Preserve the three strings and no-trimming policy. → State procedure: Apply the inclusive length boundary at four characters. → Show resulting labels: Oak is short; pine and willow are long.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Writing the methods section — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Writing the **methods section** — **evidence** trace.

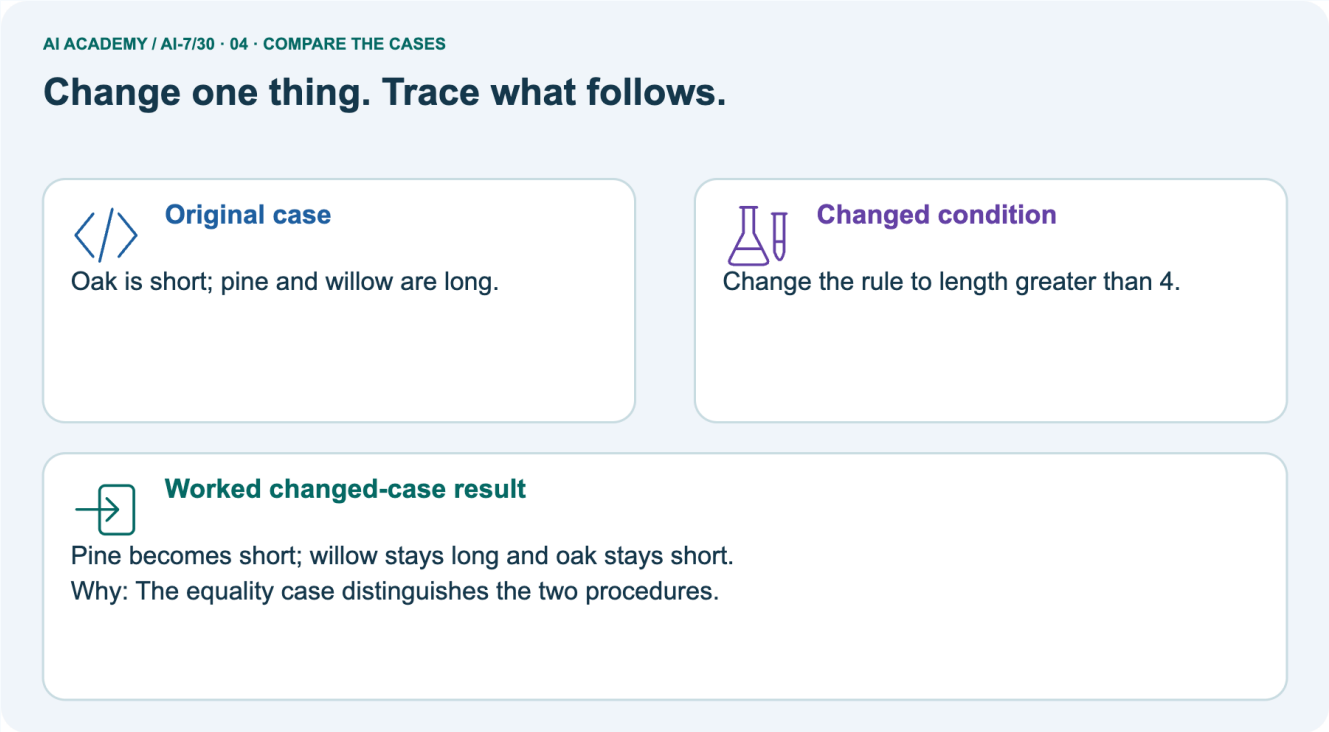
Methods says used AI on data and got good scores; code and split are absent. | Report question, units, sampling, permissions, data roles, baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions.

Distinguish prespecified and later changes. | It omits model/procedure version, data identity, units/splits, baseline, metric and scoring. Reproduction and validity cannot be assessed from the description.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

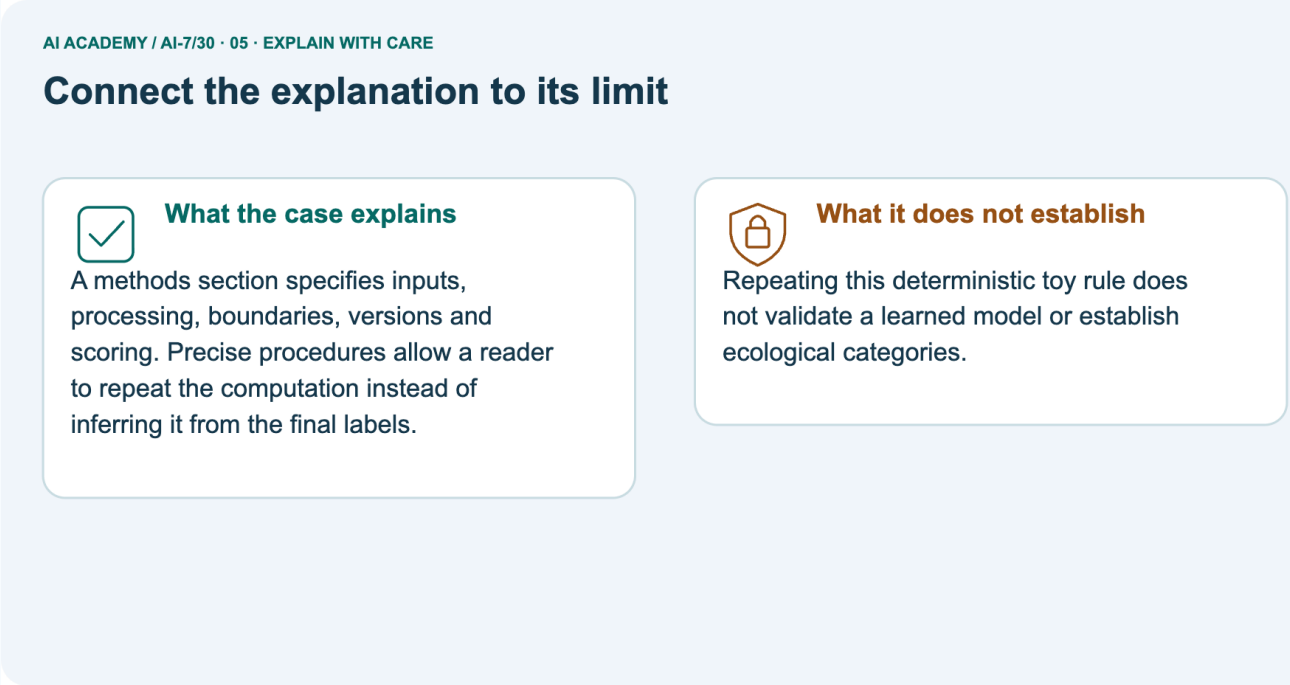
Original case: Oak is short; pine and willow are long. Changed condition: Change the rule to length greater than 4. Expected result: Pine becomes short; willow stays long and oak stays short.

Reason: The equality case distinguishes the two procedures.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

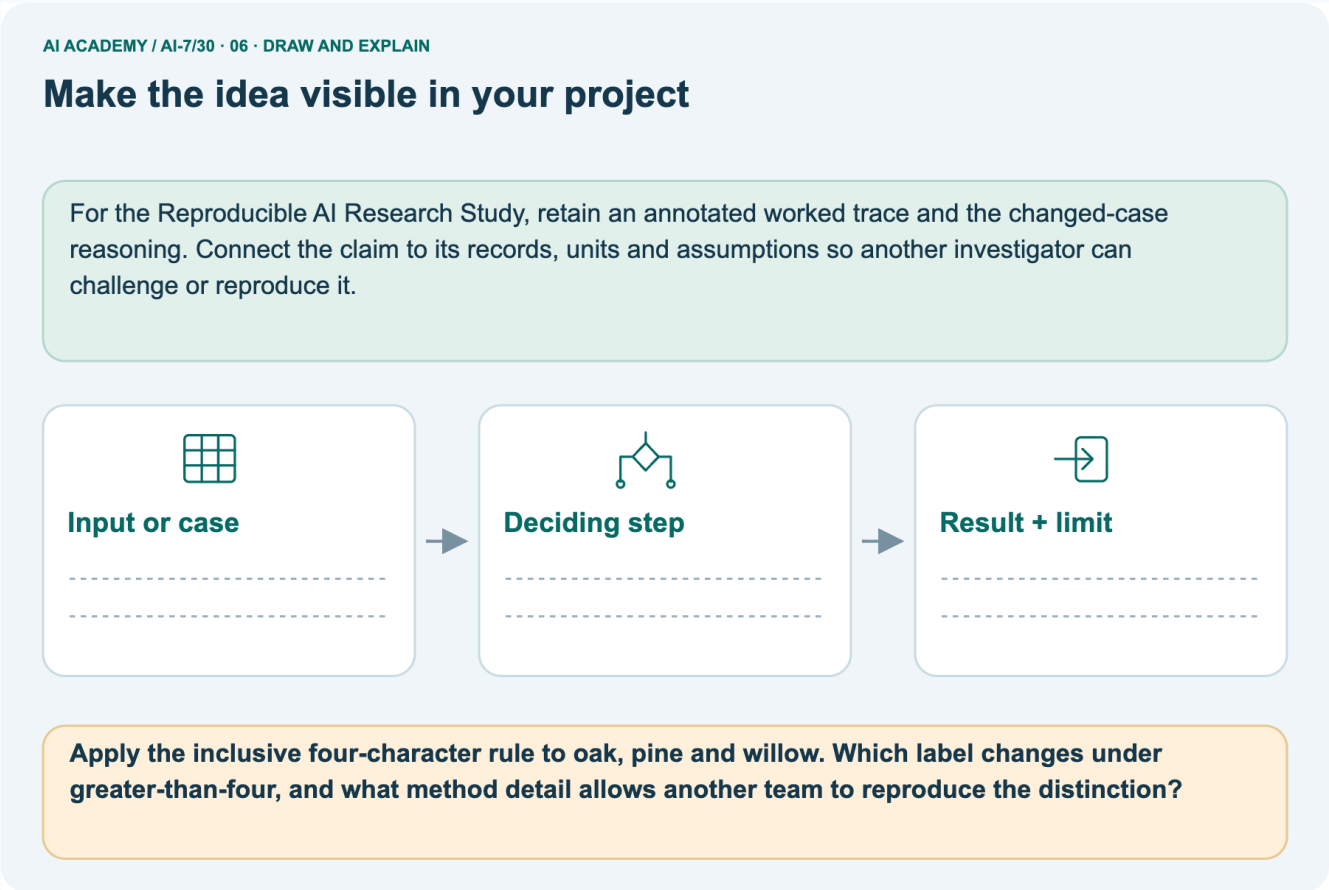
A **methods section** specifies **input**s, **process**ing, boundaries, versions and scoring. Precise procedures allow a reader to repeat the computation instead of inferring it from the final labels.

Limit: Repeating this deterministic toy rule does not validate a **learned model** or establish ecological categories.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the inclusive four-character rule to oak, pine and willow.

Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.